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Design, fabrication and mechanical properties of an auxetic cylindrical sandwich tube with improved energy absorption

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ABSTRACT

Recently, the combination of sandwich structures and auxetic structures has been extensively researched, because of its excellent mechanical properties. However, there has been limited research on sandwich structures designed as cylindrical sandwich tubes. Therefore, this paper proposes a cylindrical sandwich tube filled with elliptical perforated auxetic structure (CSTE), and uses a cylindrical sandwich filled with double-arrowed auxetic structure (CSTD) as a comparison. The results indicate that CSTE exhibits superior energy absorption capacity and higher load-bearing capacity. Additionally, the elliptical perforated auxetic core layer alters the distribution of the original energy absorption contribution rate in CSTD. Moreover, parametric analysis is conducted using the experimentally validated model to investigate the influence of various parameters on the mechanical properties of CSTE. The results show that an increase in the hollow section ratio is most effective in increasing the stiffness of CSTE, and the radius thickness ratio plays a dominant role in the energy absorption capacity. Furthermore, when the ratio of the major and minor axis of an ellipse is 1.88, the energy absorption capacity of the CSTE is significantly enhanced. These findings can facilitate the application of cylindrical sandwich tubes in protective engineering.

1. Introduction

The concept of the sandwich structure was first defined by Fairbairn in 1849, which consists of two skin layers and a core layer [1–3]. Typical sandwich structures include honeycomb sandwich structures [4,5], foam sandwich structures [6,7], composite sandwich structures [8,9], metal sandwich structures [10,11], wood sandwich structures [12,13] and paper sandwich structures [14]. Sandwich structures are characterized by their lightweight and high strength, which allows for a reduction in overall structural weight while maintaining significant strength. Thereby lowering engineering and manufacturing costs [15]. Additionally, they exhibit excellent bending resistance, high stiffness, good stability, and great energy absorption capabilities, making them widely applicable in the construction, aviation, and automotive industries [16–18].

"Auxetics" refers to negative Poisson's ratio metamaterials, which was first proposed by Evans in 1991 [19,20]. Unlike conventional structures, its outstanding deformation property is the lateral contraction (expansion) under uniaxial compression (tension) [21]. This particular deformation pattern makes auxetic materials exhibit superior mechanical properties such as shear resistance, indentation resistance, fracture resistance, energy absorption, and permeability variability [22–27]. So far, the research on auxetic materials has transitioned and expanded from 2D configuration to 3D configuration, such as perforated plate structures [28,29], re-entrant structures [30–36], rotated polygon structures [37,38], chiral structures [39–44], origami structures [45,46]. In addition, in order to obtain optimal structures with enhanced mechanical properties, bio-inspired auxetic metamaterials are gradually becoming the focus of research and receiving more and more attention [47,48]. Due to their remarkable mechanical properties, auxetic metamaterials are of great research value and are gradually explored in the fields of civil engineering [49,50], aviation [51], vehicle engineering [52], and structural protection [53].

The mechanical properties of sandwich structures are closely related to the configurations of their skin and core layers. By introducing auxetic structures, some mechanical properties of sandwich structures can be improved. Pham et al. [54] utilized auxetic honeycomb cores and functionally graded skin materials to achieve ultra-lightweight characteristics in sandwich panels. Simultaneously, research has shown that auxetic parameters have a significant impact on the free vibration

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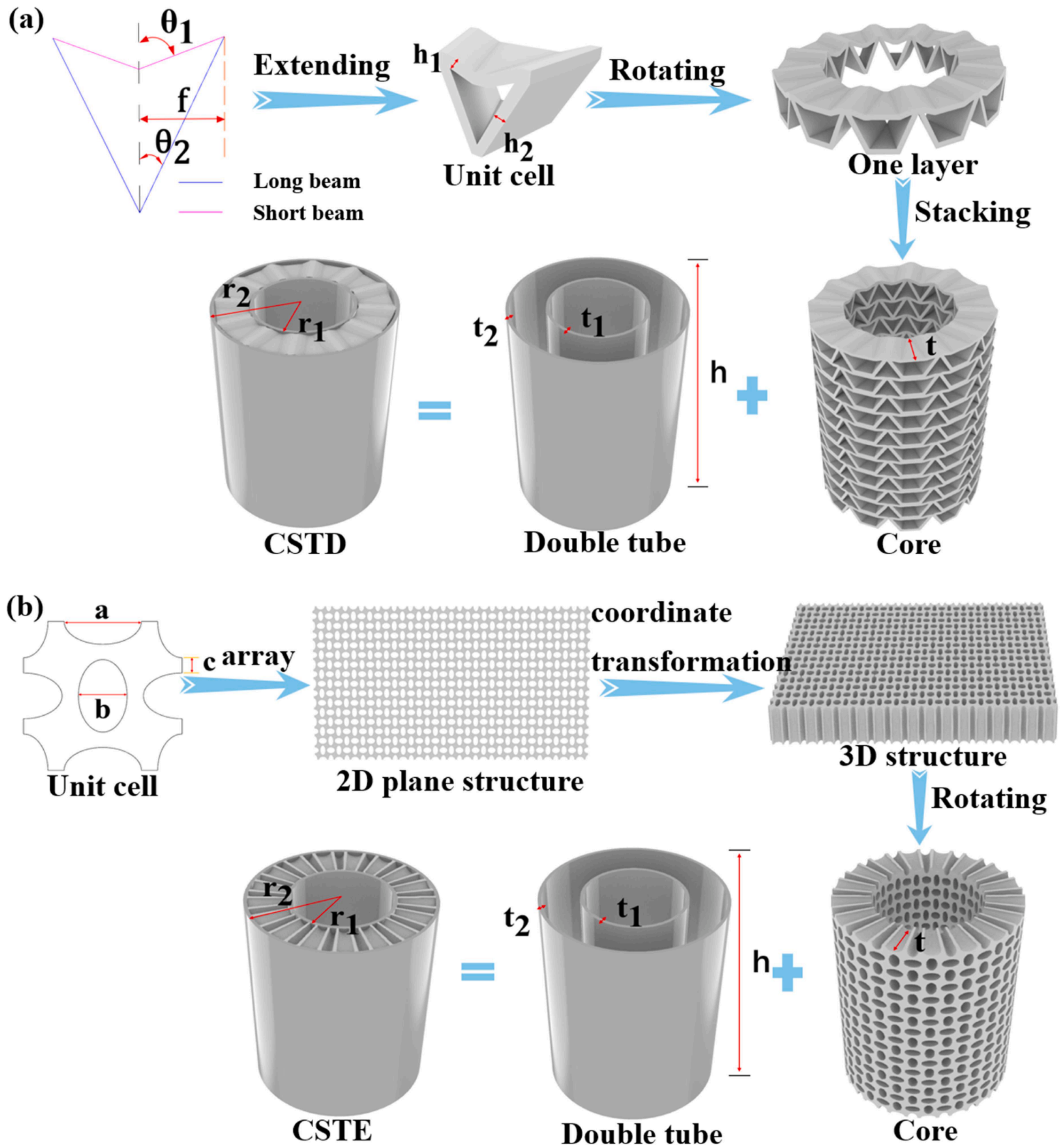

Fig. 1. Design parameters and process. (a) CSTD. (b) CSTE.

behavior of sandwich plates. Through experimental and numerical studies, Yan et al. [55] found that honeycomb sandwich panels developed using auxetic re-entrant honeycombs exhibit high damage tolerance, small deformations, and excellent blast resistance. Sengar et al. [56] first proposed findings on the influence of auxetic core parameters on the post-buckling behavior of sandwich panels under thermal loading. By increasing the ratio of core thickness to total plate thickness, the critical yield strength of the structure can be enhanced. Erkan et al. [57] discovered that by increasing the design angle in re-entrant auxetic structures, the number of failure cycles in auxetic re-entrant sandwich panels could be reduced, and their energy absorption improved. Due to the lack of research on the influence of auxeticity on the nonlinear dynamic behavior of sandwich structures, Xie et al. [58] proposed a nonlinear dynamic analysis model for sandwich beams with geometric defects, including functionally graded material faces and auxetic cores. Lu et al. [59] proposed that changing key geometric parameters and the material combination of faces and cores can significantly alter the energy absorption capacity of auxetic sandwich beams. Li et al. [60] found that reducing the absolute inclination angle of auxetic honeycomb cores can weaken the thermally induced vibration response of sandwich beams, providing a new approach to suppressing such responses. Increasing the thickness of auxetic cores can lead to increased stiffness, reduced lateral deflection, and increased critical loads in sandwich beams [61].

Based on the above content, it can be observed that the primary configurations formed by the combination of sandwich structures and auxetic structures are sandwich panels and sandwich beams, while research on sandwich tubes as a structural form is relatively scarce. In the past three years, research on the combination of sandwich tubes and auxetic structures has been as follows: Wu et al. [62] pointed out that energy absorption is caused by the deformation of sandwich structures and skins. Therefore, the failure mechanisms of the skin and the auxetic core are crucial factors in the energy absorption capacity of the structure. Zhao et al. [63] introduced a continuous density gradient into lattice cylindrical sandwich tubes, and discussed in detail the impact of the hybrid effect composed of the skin and the auxetic core on the structural energy absorption. The above studies have all explored the energy absorption effect of sandwich tubes, while the energy absorption

Table 1
Geometric parameters of cylindrical sandwich tubes.

CSTD	h_1 (mm)	h_2 (mm)	θ_1	θ_2
	1	1	69.22°	26.69°
	f (mm)	n_a	n_b	/
	2.94	11	12	
CSTE	a (mm)	b (mm)	c (mm)	n_c
	4.75	3	1	8
	n_d	/	/	/
	17	/	/	/
Public section	t_1 (mm)	t_2 (mm)	r_1 (mm)	r_2 (mm)
	1	1	15.5	27.33
	h (mm)	t (mm)	/	/
	78	10.83	/	/

contribution rate of the total energy absorption component of sandwich tubes is of great significance in optimizing structural design and improving material utilization efficiency. However, there are few studies focusing on it and its influencing factors. Therefore, the sandwich tube designed in this paper will primarily focus on studying the energy absorption contribution rate of each of its components, explaining the underlying physical mechanisms. This provides a novel approach for improving the structural design of the sandwich tube and enhancing its material usage efficiency.

In the elliptical perforated auxetic structure, the elliptical shape causes the material around the holes to deform orderly and gradually consume energy when the structure is under stress. Compared with other auxetic structures, it has a relatively uniform stress distribution during the compression process, which can reduce the local stress concentration phenomenon, guide the structure to deform orderly, and thus achieve efficient energy absorption [64,65]. Moreover, the elliptical perforated auxetic structure has structural symmetry, which makes its energy absorption relatively stable during the stress-bearing process.

Based on this, this paper proposes a new cylindrical sandwich tube filled with elliptical perforated auxetic structure (CSTE), and compares it with the cylindrical sandwich tube filled with double-arrowed auxetic structure (CSTD) [66]. The structures were fabricated using 3D printing technology with thermoplastic polyurethane (TPU) material. Static compression tests were conducted to investigate the energy absorption, load-displacement curves, deformation modes, and energy absorption contribution rate of various components within the two types of cylindrical sandwich tubes. Subsequently, a parametric analysis is performed on the CSTE to the effects of hollow section ratio, radius thickness ratio, core thickness, and the ratio of the major and minor axis of the ellipse on its mechanical properties. Furthermore, this paper proposes the envisioned applications of CSTE in soft robots. Finally, the CSTE provides a promising approach for the design of structures such as sandwich tubes, and holds broad application prospects in structural protection.

2. Design, fabrication and material properties

This section introduces the design methodologies and fabrication processes of CSTD and CSTE. Subsequently, material properties are obtained through tensile testing of standard dumbbell-shaped specimens.

2.1. Structural design

The cylindrical sandwich tube consists of an inner and outer tube with an auxetic core. This paper designs a novel cylindrical sandwich tube filled with an auxetic core based on CSTD, namely CSTE, and uses CSTD as a comparison. As shown in Fig. 1(a), this is the design process of CSTD. The double-arrowed auxetic structure is described by five independent parameters: thickness and angles of short and long beams (h_1, h_2, θ_1, θ_2), and the half-width f. The cylindrical double-arrowed auxetic structure consists of n_a layers, with each layer composed of n_b unit cells. The design process of CSTE is shown in Fig. 1(b). The elliptical perforated auxetic structure is described by three independent parameters: the major axis length a, the minor axis length b, and the spacing c of the ellipses. The cylindrical elliptical perforated auxetic structure consists of n_c layers, with each layer composed of n_d unit cells. The inner tube thickness is defined as t_1, the outer tube thickness as t_2, the inner tube radius as r_1, the outer tube radius as r_2, and the height of the tubes as h for both structures. The thickness of the core layers of CSTD and CSTE is defined as t. As shown in Table 1, all parameters for both structures are listed.

2.2. Fabrication of CSTD and CSTE

As shown in Fig. 2, both CSTD and CSTE are printed using TPU material with a density of 1.11 g/cm^3. One CSTD and one CSTE are fabricated respectively as experimental samples for study, both with a density of 1.11 g/cm^3. They are manufactured by the S-TPU 500 printer

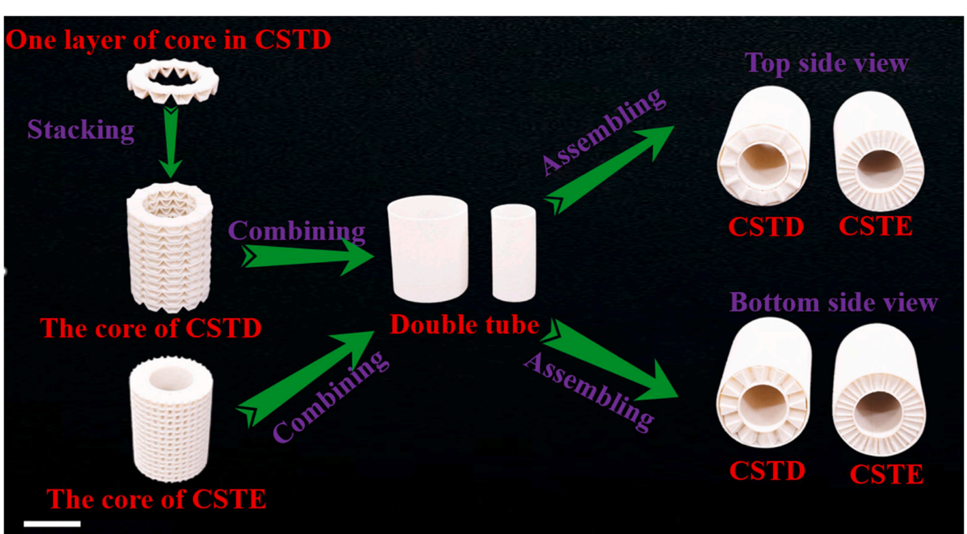

Fig. 2. 3D printing models and assembly process of CSTD and CSTE.

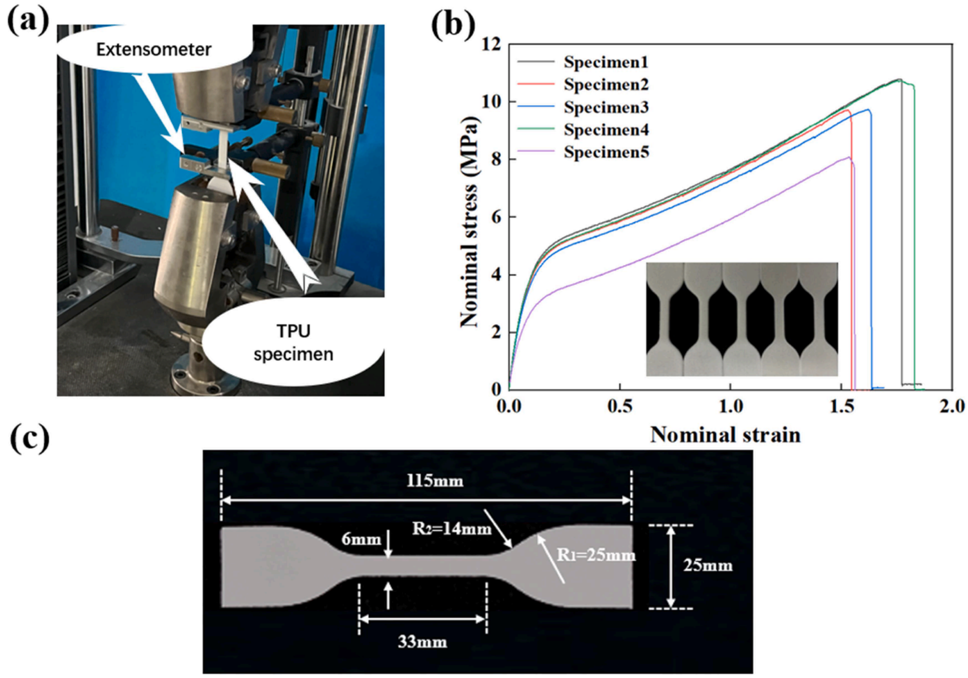

Fig. 3. Schematic illustration of the (a) Experimental setup for testing 3D-printed specimens. (b) Stress-strain curves of TPU standard dumbbell-shaped specimens. (c) Size of TPU standard dumbbell-shaped specimen.

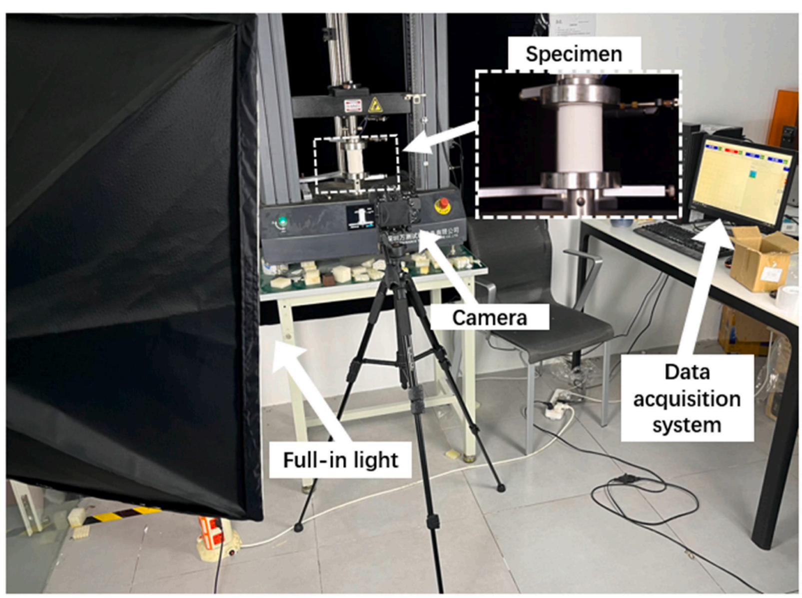

Fig. 4. Experimental setup for the compression test.

(Meizhou Xinwanwu Technology Ltd. China), a high-resolution 3D printer with a resolution range of 0.08 to 0.3 mm. Selective laser sintering with a sintering temperature of 190 °C is utilized for the printing process, resulting in essentially isotropic structures.

2.3. Material properties of TPU

The material properties of the structure were obtained through uniaxial tensile testing of standard dumbbell-shaped specimens, as exhibited in Fig. 3(a). According to the *ISO 37:2017 International standard rubber, vulcanized or thermoplastic-determination of tensile stress-strain properties standard*, five standard dumbbell-shaped tensile specimens of TPU were tested by a universal testing machine (WANCEETM-104B) at a speed of 100 mm/min. The thickness of the dumbbell-shaped specimens is 2 mm, and their specific dimensions are shown in Fig. 3(c). Fig. 3(b) shows the true stress-strain curves of five tensile specimens. Due to the error of the test conditions and the defects of the specimen, the stress-strain curve of specimen 5 is quite different from the others. Except for the failure strain, the stress-strain curves of specimens 1, 2, 3, and 4 are relatively similar. The difference in failure strain is due to the existence of random micro defects in the manufacturing process. The data collected from the specimen tests will be used for validation and analysis of the finite element (FE) model.

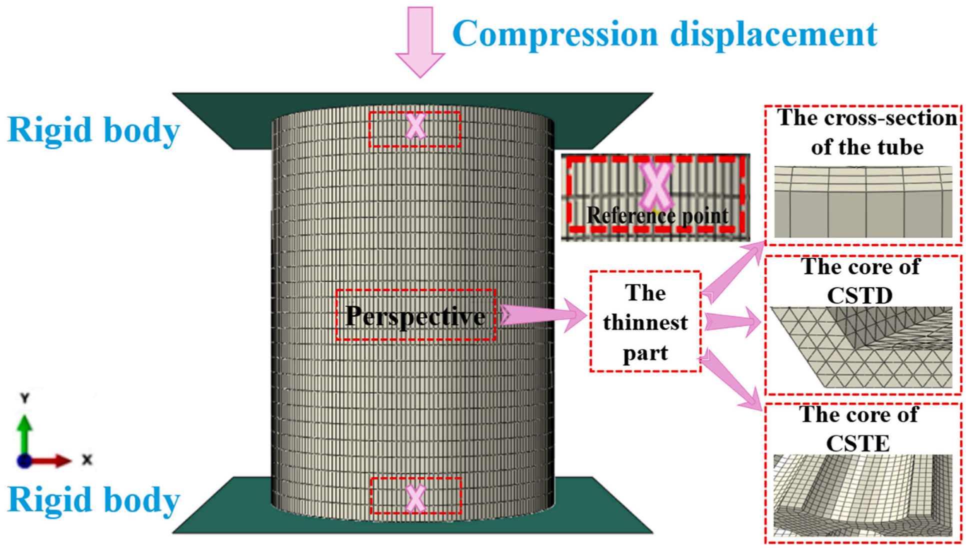

Fig. 5. FE model and mesh for the cylindrical sandwich tube.

3. Experimental setup and FE models

The experimental setup for the cylindrical sandwich tube proposed in this paper is shown in Fig. 4. The uniaxial compression test was conducted using a universal testing machine (WANCE ETM-104B, China) with a range of 10 kN and a maximum force indication error of less than ±0.5%. The compression rate for the cylindrical sandwich tube was set to 4 mm/min, corresponding to a quasi-static compression process.

The commercial software Abaqus/Explicit 6.14 was used for numerical simulation. Fig. 5 shows the FE models of the cylindrical sandwich tubes. The core layer of CSTD adopts ten-node modified quadratic tetrahedral elements (C3D10 M). The CSTE adopts eight-node linear solid elements with reduced integration (C3D8R). Considering the impact of mesh size on both computational accuracy and efficiency, a mesh size of 0.4 mm was selected during the mesh convergence analysis of the model.

The structural model in this paper is a hyperelastic model with isotropic material properties, the strain energy potential is a polynomial, the input source is a coefficient, and the strain potential energy order is 1. Dynamic explicit and general contact methods were utilized in the FE model. In terms of contact properties, the tangential friction coefficient was set to 0.3, and normal contact was set to hard contact to prevent penetration of elements within the structure. During the FE analysis process, in order to improve computational efficiency, the target time increment for mass scaling was set to 1×10^{-5}. During the compression process, the cylindrical sandwich tube was placed between two separate rigid bodies. Both the top and bottom rigid bodies are set as analytical rigid bodies, where the bottom rigid body is fixed and the top rigid body is set to undergo a downward displacement.

4. FE analysis and discussion

In this section, the results of experiments and FE simulations are compared, indicating that the numerical calculations match well with the experimental results. This validates the effectiveness of the FE model. Consequently, an in-depth study of the energy absorption capabilities of the two structures is conducted using FE analysis. Finally, the load-displacement curve responses and deformation modes of the two structures are analyzed.

4.1. Energy absorption indicators and dense points

The energy absorption capacity is evaluated by calculating four indicators [26]: total energy absorption (EA), mean crushing load (MCL), crushing load efficiency (CLE), and specific energy absorption (SEA). Their definitions are as follows:

EA refers to the amount of energy that a structure or material can absorb and convert into internal energy when subjected to an external load under specific conditions. The energy absorption capacity of the structure can be intuitively compared based on the magnitude of EA, which can be calculated from the envelope size of the load-displacement curve and the horizontal axis:

$$EA = \int_0^d F d\delta \tag{1}$$

Where F is the compression load, d is the loading displacement.

Regarding energy absorption, a higher MCL suggests better performance due to increased energy absorption during compression. MCL represents the energy absorbed per unit displacement, which is calculated as the ratio of EA to displacement:

$$MCL = \frac{EA}{d} \tag{2}$$

CLE is an important indicator for evaluating the energy absorption capacity of a structure during crushing. If CLE is close to or greater than 1, it indicates that the material or structure has excellent energy absorption capacity during crushing and can effectively resist external impacts. CLE is calculated as the ratio of MCL to peak force ($P_{\max}$):

$$CLE = \frac{MCL}{P_{\max}}$$

Comparing total energy absorption alone is unfair for structures of varying weights, due to relative density's influence. Higher specific energy absorption (SEA) values indicate better energy absorption capacity of materials during collisions. SEA is the ratio of absorbed energy to unit mass m:

$$SEA = \frac{EA}{m} \tag{4}$$

Energy absorption efficiency is generally defined as the ratio of the actual energy output to the total input energy during the process of energy conversion or transfer for a system. When less energy is required to achieve the same effect, the energy absorption efficiency is higher.

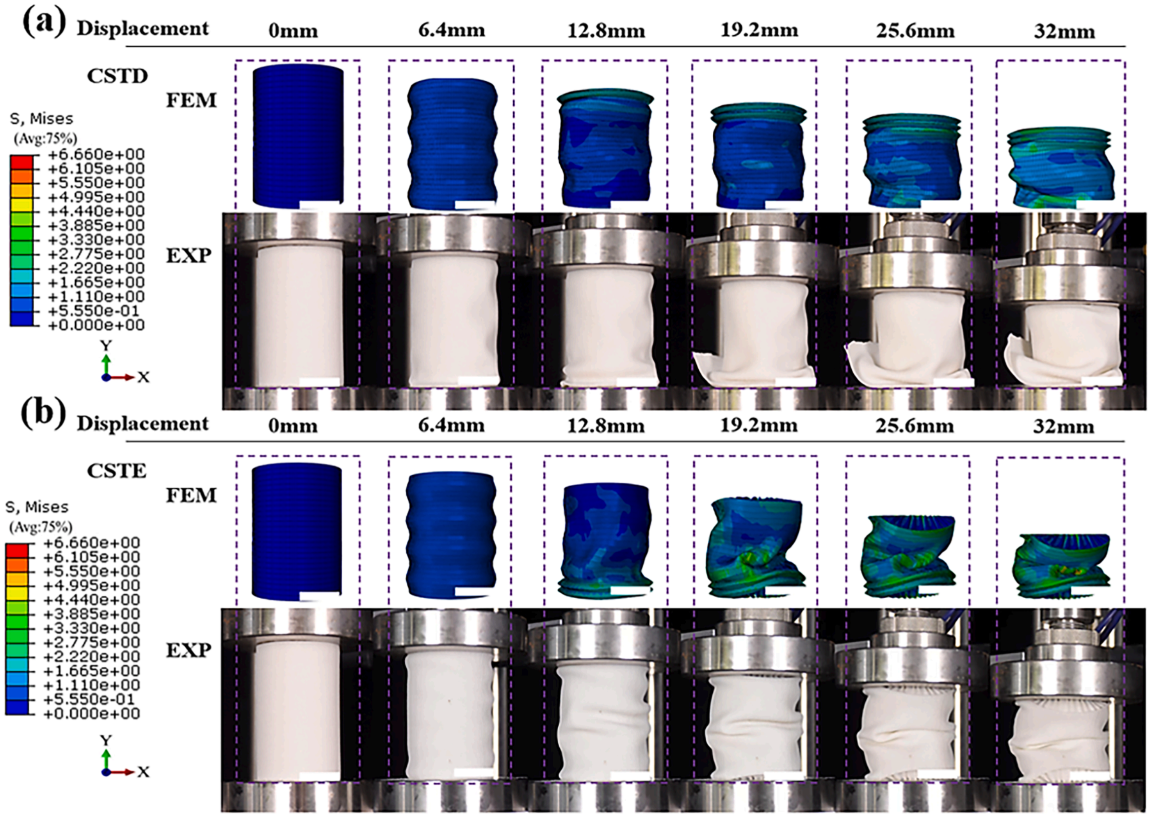

Fig. 6. Comparison of deformation modes between experiment and FE. (a) CSTD. (b) CSTE. (Scale bar: 5.5 mm).

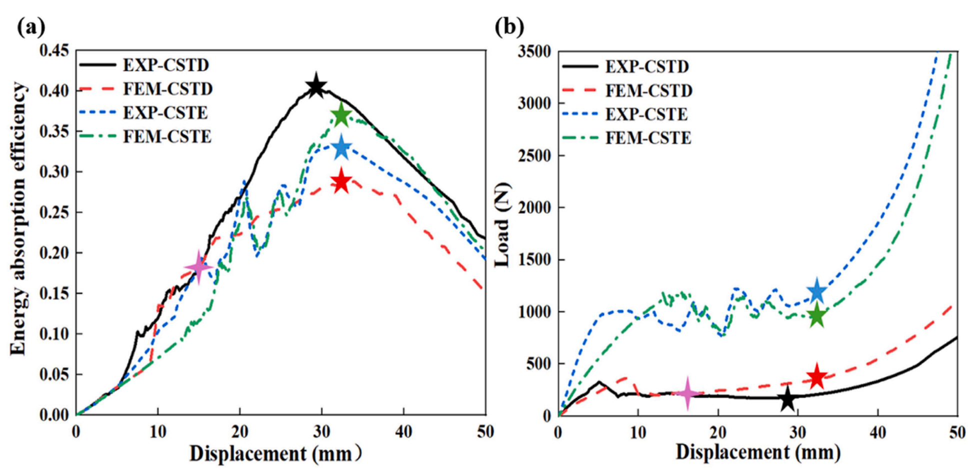

Fig. 7. Some mechanical properties of the two structures in the experiment and FE. (a) Energy absorption efficiency curves. (b) Load-displacement curves.

Conversely, if more energy is needed to attain the same effect, the energy absorption efficiency is lower. Therefore, higher energy absorption efficiency can achieve more significant energy saving effects to reduce economic costs. The energy absorption efficiency can be expressed as the integral of the stress-strain curve divided by the corresponding stress σ_a:

$$E_d(\varepsilon_a) = \frac{\int_0^{\varepsilon_a} \sigma(\varepsilon) d\varepsilon}{\sigma_a}, 0 \le \varepsilon_a < 1 \tag{5}$$

Where ε_a represents the instantaneous strain that corresponds to the stress value σ_a. The densification strain ε_d can be determined by the maximum energy absorption efficiency. Calculate EA based on the loading displacement d corresponding to the densification strain.

4.2. Verification of FE models

This section aims to validate the FE model by the experimental results, specifically focusing on the load-displacement curves and deformation processes. As depicted in Fig. 6(a), when analyzing the deformation process of CSTD, notable differences can be observed between the experimental and FE simulation results. During the experimental process, the outer tube bottom of CSTD was torn at approximately 16 mm of load displacement. This was due to defects in the material printing process. However, if we overlook this specific reason, the deformation mode observed in the experiment matches well with the FE simulation. As seen in Fig. 6(b), when considering the deformation process, the experimental and FE simulation results of CSTE

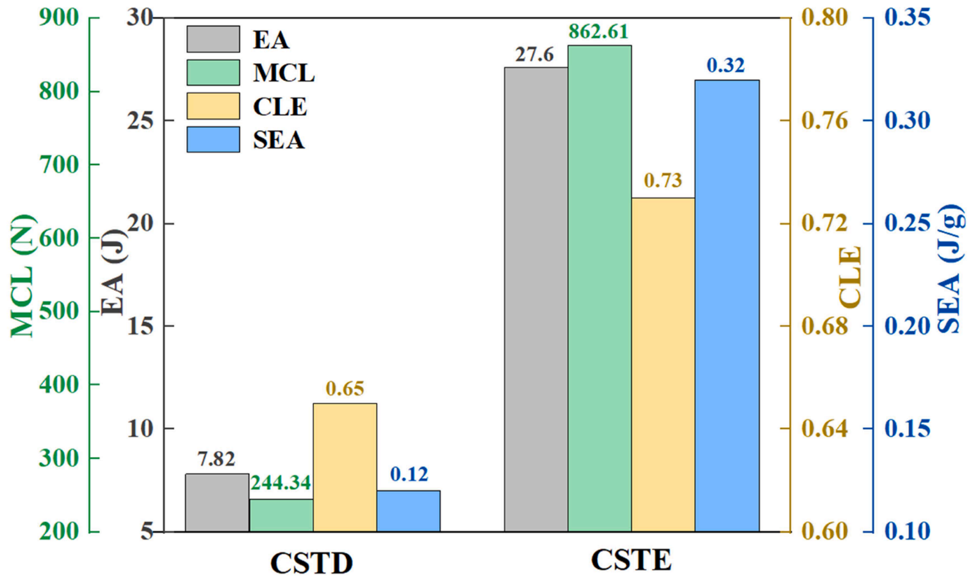

Fig. 8. Energy absorption capacity indicators of CSTD and CSTE.

demonstrate a high degree of consistency.

As shown in Fig. 7(a), the dense points of CSTD and CSTE in both experiments and FE simulations are determined based on the peaks of the energy absorption efficiency curves. These dense points are marked with five-pointed stars. Due to the tearing of the outer tube bottom of CSTD at a loading displacement of 16 mm during the experiment, the subsequent loading loads differ from those in the FE simulations. This results in a significant discrepancy between the energy absorption curves of CSTD in experiments and FE simulations, as well as different positions of the dense points. The energy absorption curves of CSTE in experiments and FE simulations showed good agreement, and the positions of the dense points are consistent.

As illustrated in Fig. 7(b), when examining the load-displacement curves, it can be seen that the trends of the experimental and FE curves for CSTD are in agreement up to a displacement of 16 mm. Nevertheless, due to minute discrepancies between 3D printed structures and FE methods, particularly in areas such as contact characteristics, material parameters, and boundary conditions, the FE simulation exhibits a slightly lower slope during the initial elastic stage. Furthermore, imperfections in the experimental setup and variations in material printing accuracy can introduce uncertainties in fracture behavior, which contribute to differences in load and peak load values between the experiment and simulation. After reaching a displacement of 16 mm, CSTD experienced a gradual decrease in load followed by an increase, due to the tearing of the outer tube bottom during the experiment. In contrast, the FE simulation operates under ideal conditions, resulting in a continuous increase in load. Fig. 7(b) presents the load-displacement curves, revealing that the trends of the experimental and FE simulations for CSTE are in agreement. Similarly, during the initial elastic stage, the slope of the FE simulation is slightly lower than that of the experiment.

In general, the errors between the experimental results and FE simulations for the two structures are within an acceptable range, validating the validity and feasibility of the FE models, subsequently, research can be conducted based on the FE models in areas such as energy absorption and deformation modes in parameterized studies.

4.3. Energy absorption analysis

Sandwich cylindrical tubes are lightweight structures with excellent load-bearing and energy absorption characteristics. The energy absorption capacity of CSTD and CSTE is compared by calculating energy absorption indicators.

Table 2
Energy absorption of each component within CSTD and CSTE.

	CSTD	CSTE
EA	7.82 J	27.6 J
EA^{tubes}	3.38 J	6.26 J
EA^{core}	3.26 J	15.81 J
$EA^{interaction}$	1.18 J	5.53 J

The dense point positions of CSTD and CSTE are calculated through the energy absorption efficiency curves. In the FE simulation, the dense points of CSTD and CSTE are both at 32 mm. The position of the dense point can serve as a metric for the usable stroke length of EA. The indicators of the specimens in the FE analysis are shown in Fig. 8. The results show that all indicators of CSTE are higher than those of CSTD. The EA of CSTE is 3.5 times that of CSTD, the SEA of CSTE is 2.7 times greater than that of CSTD, and its MCL and CLE are also higher than those of CSTD. Overall, based on various indicators, the energy absorption effect of CSTE is better than that of CSTD.

The total energy absorption (EA) of the sandwich cylindrical tube can be divided into three parts:

$$EA = EA^{tubes} + EA^{core} + EA^{interaction} \tag{6}$$

Where the EA^{tubes} is the energy absorbed by the inner and outer cylindrical tubes, the EA^{core} is the energy absorbed by the middle auxetic core layer, and the $EA^{interaction}$ is the energy dissipated by the interaction between the inner and outer cylindrical tubes and the middle core layer. The energy absorption of each part of the sandwich cylindrical tubes is calculated using FE analysis. As shown in Table 2, the EA^{tubes} of CSTD and CSTE accounts for 43% and 23% of their EA, respectively. The EA^{core} of CSTD and CSTE absorb 42% and 57% of their EA, respectively. The $EA^{interaction}$ of CSTD and CSTE accounts for 15% and 20% of their EA, respectively. The formula for the energy absorption contribution rate (EACR) of a cylindrical sandwich tube is as follows:

$$EACR^{part} = \frac{EA^{part}}{EA} \tag{7}$$

From the above data, it can be seen that in CSTD, the $EACR^{tubes}$ is the largest, while in CSTE, the $EACR^{core}$is the largest.

The reason why the energy absorption effect of CSTE is better than

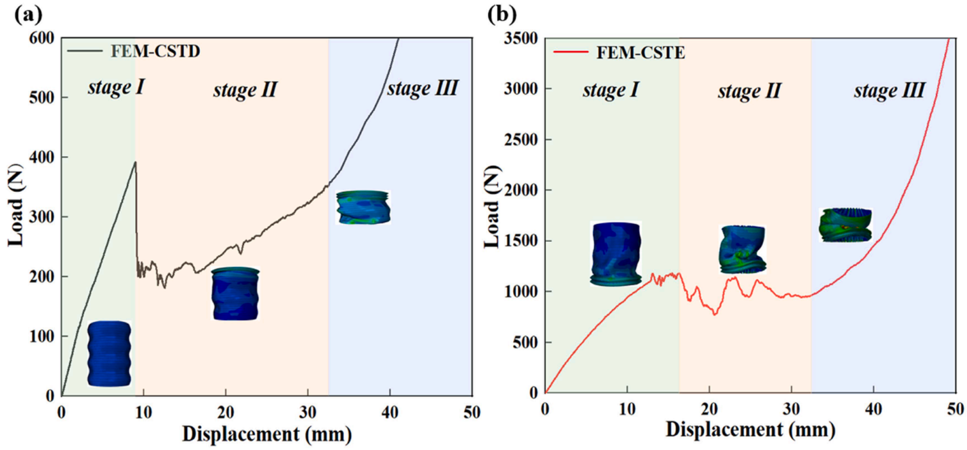

Fig. 9. Load-displacement curves in FE. (a) CSTD. (b) CSTE.

that of CSTD is that its elliptical perforated auxetic core layer structure has excellent energy absorption properties. It changes the original distribution of EACR in CSTD, shifting from the inner and outer tubes being the dominant contributors to the auxetic core layer. This can reduce the damage to the inner and outer thin-walled tubes during stress to a certain extent.

Compared with the elliptical perforated auxetic structure, the double-arrowed auxetic structure has relatively weaker overall connectivity and support. When subjected to external loads, it cannot effectively transfer and distribute the pressure to various parts of the core layer. This structural characteristic causes the double-arrowed auxetic core layer to be unable to fully exert its energy absorption potential. Moreover, since the inner and outer tubes, as relatively continuous and complete structures, can better withstand and disperse external pressure and resist deformation, the inner and outer tubes undertake the main energy absorption role. When the elliptical perforated auxetic structure is under pressure, the elliptical holes provide the structure with more degrees of freedom for deformation, enabling the core layer to deform and absorb energy more effectively. At the same time, the elliptical perforated structure can disperse energy to a larger range of the core layer through hole deformation and local buckling of the structure. Compared with the double-arrowed auxetic structure, the elliptical perforated auxetic structure can form a more effective stress transfer and energy dissipation path in the core layer, enabling the core layer to assume the main energy absorption role.

4.4. Load-displacement curves and deformation modes

The sandwich cylindrical tube undergoes three stages during compression: initial elastic, plateau, and densification. Its deformation modes can be classified as stable fold mode type I; unstable local buckling mode type II; transverse shear mode type III; and middle collapse mode type IV.

As shown in Fig. 9(a), the initial elastic stage of CSTD occurs when the displacement is between 0 and 9 mm. As shown in Fig. 9(b), When the displacement is between 0 and 16 mm, the initial elastic stage of CSTE occurs. During this stage, the load increases rapidly with displacement until reaching the initial peak load. It can be observed from the figure that the initial peak load of CSTE is much higher than that of CSTD. Structures with higher initial peak load can withstand greater external load without yielding or failing, and are less prone to instability. They also have a longer service life, reducing maintenance and replacement frequency. Additionally, the elastic modulus of CSTE is higher, indicating greater stiffness. It deforms less when subjected to load, better maintaining the shape and size of the structure. It can also effectively suppress vibration propagation, enhancing the stability and safety of the structure. At the end of this stage, local buckling occurs in a certain part of both structures, causing the load to decrease rapidly after reaching the initial peak. The deformation mode of CSTD during the initial elastic stage is type I, where the tube walls of the structure form many regular small folds. The deformation mode of CSTE during the initial elastic stage is a combination of type I and type II, and the structure undergoes local buckling at the end of this stage.

The plateau stage of CSTD occurs when the displacement is between 9 and 32 mm as shown in Fig. 9(a). As shown in Fig. 9(b), the plateau stage of CSTE occurs when the displacement is between 16 and 32 mm. During the plateau stage, as the displacement increases, the load fluctuates around the average load. The formation of folds during this compression process leads to the fluctuation of the load. It can be observed from the figure that CSTE has a higher plateau stress compared to CSTD, indicating its better energy absorption capability and good load-bearing capacity. The deformation mode of CSTD during the plateau stage is a combination of type I and type II, with folds forming on the tube wall and local buckling occurring in the structure. The deformation mode of CSTE during the plateau stage is a combination of type II and type IV. In the early stage of this stage, local buckling occurs and the tube wall is accompanied by minor folds. Subsequently, the middle part collapses and divides into two slope sections.

As shown in Fig. 9, the densification stages of both CSTD and CSTE occur after a displacement of 32mm. During the densification stage, due to densification, the load increases significantly with the increase in displacement. The deformation mode of CSTD and CSTE during the densification stage is mainly type IV.

Table 3
Geometric parameters of CSTE with different hollow section ratios.

μ	a (mm)	b (mm)	c (mm)	t_1 (mm)	t_2 (mm)	r_1 (mm)	r_2 (mm)	t (mm)	h (mm)
0.589	4.75	3	1	1	1	15.5	27.33	10.83	78
0.600	4.75	3	1	1.5	1	15.5	27.33	10.83	78
0.612	4.75	3	1	2	1	15.5	27.33	10.83	78
0.624	4.75	3	1	2.5	1	15.5	27.33	10.83	78

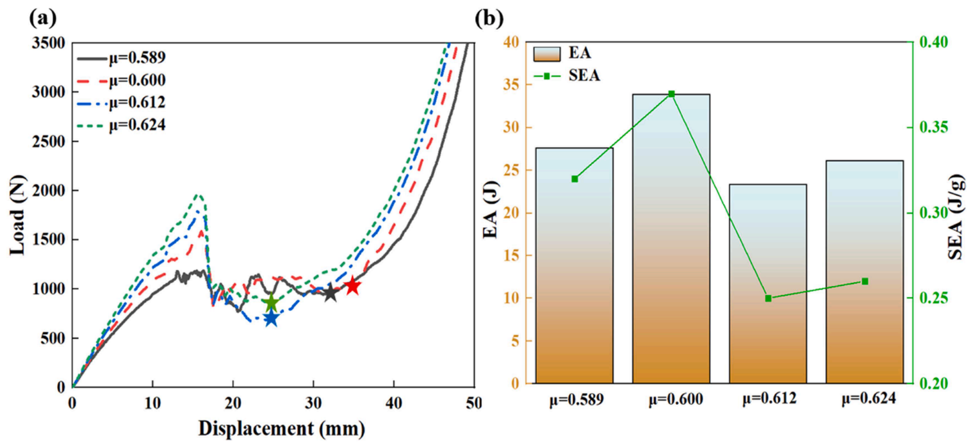

Fig. 10. Comparison of CSTE with different hollow section ratios. (a) Load-displacement curves. (b) The EA histogram and SEA line curve.

Table 4
Energy absorption of various components in CSTE with different hollow section ratios.

μ	0.589	0.600	0.612	0.624
EA	27.6 J	33.86 J	23.34 J	26.16 J
EA^{tubes}	6.26 J	7.15 J	6.1 J	7.85 J
EA^{core}	15.81 J	16.86 J	10.51 J	11.82 J
$EA^{interaction}$	5.53 J	9.85 J	6.73 J	6.49 J

5. Parametric studies

Through comparative analysis between experiments and FE simulations, the FE simulations show good agreement with the experimental results. Furthermore, FE simulations analyzed energy absorption, load-displacement response, and deformation modes of CSTD and CSTE. Results indicate that CSTE outperforms CSTD in energy absorption and other mechanical properties. As a result, this section further investigates CSTE using FE analysis, exploring the effects of hollow section ratio, radius thickness ratio, core layer thickness, and the ratio of the major and minor axis of the ellipse on structural energy absorption and other related aspects.

5.1. Effect of hollow section ratio

In this section, the hollow section ratio (μ) is introduced as a parameter to describe the influence of the inner tube thickness on CSTE. The design dimensions of the structure are shown in Table 3. The hollow section ratio of CSTE is defined as:

$$\mu = \frac{2r_1}{2r_2 - 2t_1} \tag{8}$$

Where r_1 is the inner tube radius, r_2 is the outer tube radius, and t_1 is the inner tube thickness.

As shown in Fig. 10(a), the initial load increases more and more rapidly as the hollow section ratio increases and there is a positive correlation between the magnitude of the initial peak load of the structure and the magnitude of the hollow section ratio. Among the four structures, the one with μ=0.600 has the highest platform stress, while the one with μ=0.612 has the lowest. Both μ=0.589 and μ=0.600 exhibit longer stress plateaus. As shown in Fig. 10(b), the values of EA and SEA of CSTE have a nonlinear relationship with its hollow section ratio. When μ=0.6, the EA and SEA of CSTE reach their maximum, which is 33.86 J and 0.37J/g respectively, indicating the best energy absorption capacity.

As exhibited in Table 4, with the increase in the hollow section ratio, the $EACR^{tubes}$ and $EACR^{interaction}$increases while the $EACR^{core}$ decreases. Within the range of μ=0.589 to μ=0.624 for CSTE, the $EACR^{tubes}$increases from 23% to 30%, and the $EACR^{interaction}$increases from 20% to 25%. While the $EACR^{core}$decreases from 57% to 45%.

5.2. Effect of radius thickness ratio

In this section, the effect of the thickness of the outer tube and outer diameter on CSTE is described by introducing the radius thickness ratio (k) as a parameter. The design dimensions of the structure are shown in Table 5. The radius thickness ratio is defined as:

$$k = \frac{2r_2}{t_2} \tag{9}$$

where r_2 is the outer tube radius and t_2 is the outer tube thickness.

As shown in Fig. 11(a), as the ratio of radius thickness ratio decreases, the duration of the initial elastic stage of the load-displacement curve becomes shorter. CSTE with a smaller radius thickness ratio can reach the initial peak load in a shorter time. Compared with parametric studies focusing on the hollow section ratio, the overall slope during the initial elastic stage is lower. Therefore, an increase in the hollow section ratio has a greater impact on improving the stiffness of the overall structure. Compared with the other three structures, the CSTE with $k =$ 23.06 has a higher platform stress, with smaller load fluctuations. Therefore, when the radius thickness ratio is low, the energy absorption of CSTE is more stable. As shown in Fig. 11(b), There is a nonlinear relationship between the EA and SEA values of CSTE and its radius thickness ratio. Compared to changes in the hollow section ratio

Table 5
Geometric parameters of CSTE with different radius thickness ratios.

k	a (mm)	b (mm)	c (mm)	t_1 (mm)	t_2 (mm)	r_1 (mm)	r_2 (mm)	t (mm)	h (mm)
54.66	4.75	3	1	1	1	15.5	27.33	10.83	78
37.11	4.75	3	1	1	1.5	15.5	27.83	10.83	78
28.33	4.75	3	1	1	2	15.5	28.33	10.83	78
23.06	4.75	3	1	1	2.5	15.5	28.83	10.83	78

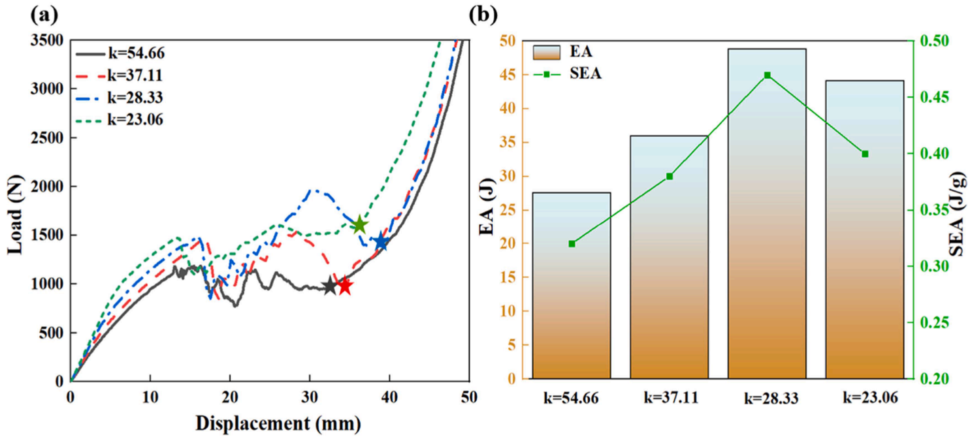

Fig. 11. Comparison of CSTE with different radius thickness ratios. (a) Load-displacement curves. (b) The EA histogram and SEA line curve.

Table 6
Energy absorption of various components in CSTE with different radius thickness ratios.

k	54.66	37.11	28.33	23.06
EA	27.6 J	35.93 J	48.8 J	44.09 J
EA^{tubes}	6.26 J	9.11 J	12.74 J	13.99 J
EA^{core}	15.81 J	18.3 J	24.31 J	22.83 J
$EA^{interaction}$	5.53 J	8.52 J	11.75 J	7.27 J

discussed in the previous section, for each k value, the EA and SEA are higher than those corresponding to the μ value, indicating that changing the radius thickness ratio is more effective in improving energy absorption.

As exhibited in Table 6, within the range of $k = 54.66$ to $k = 23.06$ for CSTE, the $EACR^{tubes}$ increases from 23% to 32%, while the $EACR^{core}$ decreases from 57% to 52%, and the $EACR^{interaction}$ decreases from 20% to 16%. Compared to the impact of changes in the hollow section ratio on EACR in the previous section, it can be observed that the hollow section ratio has a greater influence on $EACR^{core}$, while the radius thickness ratio has a greater influence on $EACR^{tubes}$.

5.3. Effect of core thickness

The core layer is an important component of the cylindrical sandwich tube, and it has a significant impact on energy absorption and other aspects of the overall structure. Therefore, in this section, the core thickness (t) is used as an independent variable parameter to describe its influence on CSTE. In addition, the effect of parameter μ on the structural properties is discussed due to the coupling relations between t and μ. The design dimensions of the structure are shown in Table 7, in which the thickness of the core layer is negatively correlated with the hollow section ratio.

As shown in Fig. 12(a), with the increase in core thickness, the initial peak load increases. By observing the changes in initial peak load in Fig. 10(a) and 11 (a), it is evident that the radius thickness ratio has the least influence on changes in initial peak load. In contrast, the core layer thickness exerts a moderate effect, while the hollow section ratio has the

Table 7
Geometric parameters of CSTE with different core thicknesses.

t (mm)	μ	a (mm)	b (mm)	c (mm)	t_1 (mm)	t_2 (mm)	r_1 (mm)	r_2 (mm)	h (mm)
10.83	0.557	4.75	3	1	1	1	15.5	27.33	78
11.33	0.539	4.75	3	1	1	1	15	27.33	78
11.83	0.521	4.75	3	1	1	1	14.5	27.33	78
12.33	0.503	4.75	3	1	1	1	14	27.33	78

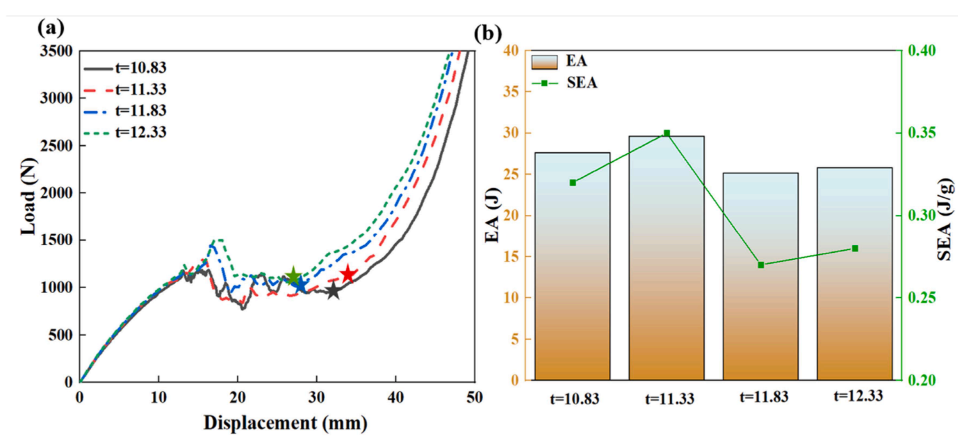

Fig. 12. Comparison of CSTE with different core thicknesses. (a) Load-displacement curves. (b) The EA histogram and SEA line curve.

Table 8
Energy absorption of various components in CSTE with different core thicknesses.

t	10.83	11.33	11.83	12.33
EA	27.6 J	29.61 J	25.14 J	25.8 J
EA^{tubes}	6.26 J	6.39 J	5.74 J	6.03 J
EA^{core}	15.81 J	18.5 J	14.83 J	15.22 J
$EA^{interaction}$	5.53 J	4.72 J	4.57 J	4.55 J

most significant impact. According to the data on changes in the hollow section ratio caused by core thickness in this section, the reason for the slow increase in initial peak load with increasing core thickness is the decrease in the hollow section ratio. In the initial elastic stage, the slope of the CSTE for different core thicknesses remains constant, so the stiffness of the CSTE is independent of the core thickness. CSTE with thicker core layers has higher platform stress, but its stress platform is shorter. As shown in Fig. 12(b), the values of EA and SEA of CSTE have a nonlinear relationship with its core layer thickness. Compared to changes in the hollow section ratio and radius thickness ratio discussed in the previous sections, it is still the most effective way to improve energy absorption capacity by changing the radius thickness ratio.

As shown in Table 8, with the increase in core thickness, the $EACR^{tubes}$ remains stable at 23%, while the $EACR^{core}$ increases slowly, from 57% to 59%. The $EACR^{interaction}$ also decreases slowly, from 20% to 18%. By observing the influence of previous parameter changes on EACR, it can be found that core thickness has a relatively small impact on EACR among different parts of the structure.

5.4. Effect of the ratio of the major and minor axis of the ellipse

The ratio of the major and minor axis of the elliptical perforated sandwich layer has a significant impact on the compressive stiffness of the sandwich tube structure, thereby affecting its energy absorption capacity. Therefore, the ratio of the major and minor axis of ellipse M(a/b) is adopted to investigate its influence on CSTE. The design dimensions of the structure are shown in Table 9.

As shown in Fig. 13(a), in the initial elastic stage, the elastic modulus of CSTE decreases as M increases, and its initial peak load also decreases accordingly. When $M = 1.58$, the plateau stress of CSTE is the largest, followed by $M = 1.88$, and finally $M = 2.4$. When M is greater than 1, the plateau stress of CSTE is negatively correlated with the value of M. The CSTE with $M = 1.88$ has the longest stress plateau. For CSTE with $M = 1.88$, the load increases sharply at a displacement of approximately 25 mm due to the auxetic contraction, which indicates the reaction load sharply increased in the plateau stage. Between 25 mm and 35 mm, stable wrinkles appear on CSTE. After 35 mm, local buckling occurs on CSTE, resulting in a sharp decrease in load. As shown in Fig. 13(b), the EA and SEA of CSTE at $M = 1.88$ is the largest, far exceeding the other three structures, which are 52.96 J and 0.61J/g respectively.

As shown in Table 10, when M=1.88, the $EACR^{tubes}$ of CSTE is 19%, the $EACR^{core}$ is 49%, and the $EACR^{interaction}$ is 32%. Compared with the other three structures, the $EACR^{core}$is significantly reduced, while the $EACR^{interaction}$ is greatly increased. The $EACR^{tubes}$ remains stable.

6. Envisioned applications of 3D auxetic cylindrical sandwich tube CSTE

In recent years, soft robots have received widespread attention due to their excellent deformability and high adaptability [67,68]. These characteristics have led to extensive research in fields such as medical treatment and exploration [69–71].

In practical use scenarios, soft robots such as prosthetic arms and legs and rehabilitation robots, face various complex situations. They need to possess sufficient safety and stability while also being lightweight for

Table 10
Energy absorption of various components in CSTE with different ratios of major and minor axis of ellipse.

M	1	1.58	1.88	2.4
EA	21.17 J	27.6 J	52.96 J	28.94 J
EA^{tubes}	3.84 J	6.26 J	10.12 J	6.18 J
EA^{core}	13.67 J	15.81 J	25.99 J	16 J
$EA^{interaction}$	3.66 J	5.53 J	16.85 J	6.76 J

Table 9
Geometric parameters of CSTE with different ratios of major and minor axis of ellipse.

M	a (mm)	b (mm)	c (mm)	t_1 (mm)	t_2 (mm)	r_1 (mm)	r_2 (mm)	t (mm)	h (mm)
1	3.75	3.75	1.125	1	1	15.5	27.33	10.83	78
1.58	4.75	3	1	1	1	15.5	27.33	10.83	78
1.88	5.25	2.8	0.85	1	1	15.5	27.33	10.83	78
2.4	6	2.5	0.625	1	1	15.5	27.33	10.83	78

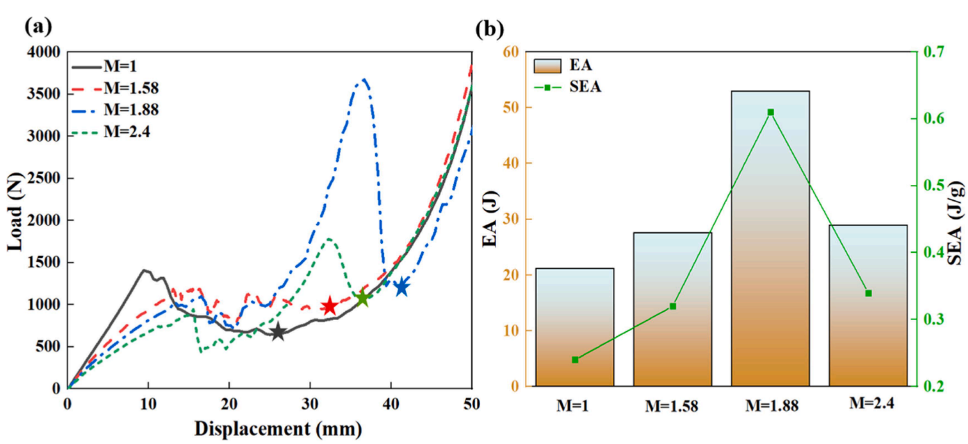

Fig. 13. Comparison of CSTE with different ratios of the major and minor axis of the ellipse. (a) Load-displacement curves. (b) The EA histogram and SEA line curve.

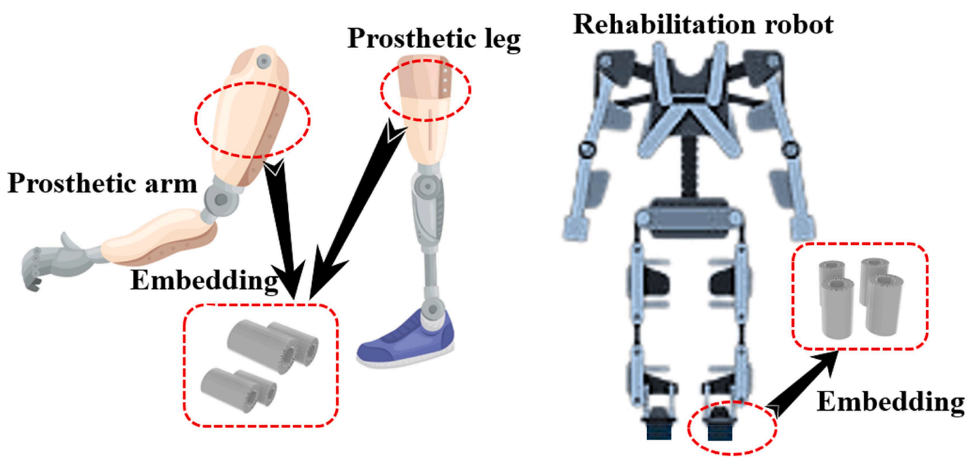

Fig. 14. Envisioned applications of the CSTE.

easy user operation. When a prosthetic arm assists a user in daily activities and accidentally encounters a collision, or when the bottom of a rehabilitation robot is subjected to external forces while assisting a patient in walking, good energy absorption performance becomes crucial. The research presented in this paper shows that the 3D-printed TPU structure CSTE has good energy absorption effect and is lightweight characteristic. This significant advantage makes it an ideal choice for protective devices on soft robots. As shown in Fig. 14, CSTE can be applied to prosthetic arms, prosthetic legs, and the bottom of rehabilitation robots through embedding. CSTE can effectively buffer the impact of these external forces with their excellent energy absorption performance, preventing damage to the relevant parts of soft robots. Additionally, they are lightweight enough to avoid excessive weight in soft robots, and provide an idea for the lightweight design of soft robot protective devices.

7. Conclusion

In this study, a cylindrical sandwich tube with an elliptic perforated auxetic structure (CSTE) is proposed, and it is compared with a cylindrical sandwich tube with a double arrowed auxetic structure (CSTD). Their energy absorption, load-displacement curves, and deformation modes are investigated. Furthermore, the experimentally verified model is used to conduct parametric analysis of CSTE from four aspects: hollow section ratio, radius thickness ratio, core layer thickness, and the ratio of the major and minor axis of the ellipse. Finally, the envisioned applications of CSTE in soft robots are proposed. Based on this work, the following conclusions can be drawn:

(1) The EA of CSTE is 3.5 times that of CSTD, the SEA of CSTE is 2.7 times that of CSTD, and its MCL and CLE are also higher than CSTD. Therefore, the energy absorption effect of CSTE is better than CSTD.
(2) In CSTD, the energy absorption is primarily contributed by the inner and outer tubes, whereas in CSTE, it is mainly attributed to the auxetic core layer. This allows the cylindrical sandwich tube to avoid damage to the inner and outer thin-walled tubes when subjected to load.
(3) The initial peak load and elastic modulus of CSTE are higher than those of CSTD. Additionally, CSTE exhibits a higher platform stress compared to CSTD. Compared to CSTD, CSTE is more suitable for conditions with large external loads, high energy absorption and stiffness requirements, and also has a longer service life.
(4) Among the three factors of hollow section ratio, radius thickness ratio, and core thickness, the radius thickness ratio has the greatest impact on the energy absorption capacity of CSTE.
(5) CSTE with a larger hollow section ratio exhibits a longer stress platform, while CSTE with a larger radius thickness ratio exhibits a shorter stress platform with lower platform stress. The increase in the hollow section ratio is most effective in improving the stiffness of CSTE.
(6) When $M = 1.88$, the $EACR^{core}$ in CSTE significantly decreases, while the $EACR^{interaction}$ significantly increases, resulting in a significant enhancement of the overall energy absorption capacity. This not only enhances the energy absorption capacity due to the increased energy dissipated through interaction but also reduces damage to the core layer and tubes of CSTE under loading, thereby extending their lifespan.

In general, CSTE exhibits excellent performance in terms of energy absorption and load-bearing capacity. Due to its lightweight material, CSTE has great potential in fields such as medical devices, automotive engineering, sports protection, and construction engineering. However, in this work, only the energy absorption of CSTE under quasi-static compression was explored. Therefore, the energy absorption of the designed CSTE under impact loading deserves further investigation in future work.

CRediT authorship contribution statement

Xiao Ji: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Yi Zhang:** Writing – review & editing, Investigation. **Wei Zhong Jiang:** Writing – review & editing. **Jun Wen Shi:** Writing – review & editing. **Yi Chao Qu:** Writing – review & editing. **Meng Li Xue:** Writing – review & editing. **Xin Ren:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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